PROJECT REPORT

On

Student Grade Predictor

Submitted in Partial Fulfillment of Award of

BACHELOR OF TECHNOLOGY

In

Computer Science and Engineering

By

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ALLIANCE SCHOOL OF ADVANCED COMPUTING

BACHELOR OF TECHNOLOGY

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CERTIFICATE

This is to certify that the project work entitled Student Grade Predictor submitted by Deepak Sanjai [2411021060141], K. Thejash [2411021060018], K. Charan Tej [2411021060059], T.k.lalith kumar [2411021060065] in partial fulfillment for the award of the degree of Bachelor of Technology Computer Science and Engineering of Alliance University, is a Bonafide work accomplished under our supervision and guidance during the academic year 2025-2026. This thesis report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

Dr R C Karpagalakshmi
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DECLARATION

We hereby declare that the project entitled Student Grade Predictor submitted by us in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (CSE) of ASAC, Alliance University, is a record of our work carried under the supervision and guidance of Dr R C Karpagalakshmi

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty guide.

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TABLE OF CONTENTS

1. INTRODUCTION
2. LITERATURE SURVEY
3. SYSTEM DESIGN
4. SYSTEM IMPLEMENTATION
5. RESULTS AND DISCUSSION
6. TESTING
7. CONCLUSION AND FUTURE ENHANCEMENTS
8. RFERENCES & BIBLIOGRAPHY

STUDENT GRADE PREDICTOR USING AI & ML

1. INTRODUCTION

In recent years, the integration of Artificial Intelligence and Machine Learning in education has opened new possibilities for improving student outcomes. One such application is the **Student Grade Predictor**, a system designed to estimate a student's final academic performance based on key factors such as study hours, attendance, and previous marks. Predicting student performance in advance helps both students and educators take timely actions to improve results and avoid academic failure.

Traditionally, teachers rely on manual estimation or experience to predict student performance, which may not always be accurate. This project addresses that limitation by introducing a data-driven approach. By analyzing historical student data, the system can identify patterns and relationships between input factors and final marks. The prediction process is fast, simple, and efficient, making it suitable for real-time use.

This system is particularly useful for:

- Early identification of weak students
- Providing timely guidance for improvement
- Supporting academic planning and decision-making

2. LITERATURE SURVEY

Student performance prediction has been an important area of research in Educational Data Mining. Earlier approaches mainly depended on manual evaluation methods or basic statistical techniques. While these methods provided some insights, they lacked accuracy and scalability.

Recent advancements in Machine Learning have enabled more efficient prediction systems. Algorithms such as Linear Regression, Decision Trees, and Neural Networks are widely used to analyze student data and predict outcomes. However, many existing systems still face challenges such as limited input factors, lack of real-time prediction, and complexity in implementation.

The major limitations of traditional and existing systems include:

- Inaccurate manual predictions
- Lack of integration of multiple influencing factors

- Time-consuming analysis
- No automated or real-time results

Therefore, there is a need for a simple, efficient, and automated system that can combine important parameters like study hours, attendance, and previous marks to provide accurate predictions.

3. SYSTEM DESIGN

The Student Grade Predictor is designed using a structured approach that consists of three main layers: input, processing, and output. The input layer collects relevant student data, the processing layer performs analysis and prediction using a machine learning model, and the output layer displays the predicted results.

The overall system follows a simple flow:
Input → Processing → Prediction → Output

The input layer includes:
- Study Hours (daily preparation time)
- Attendance (percentage of classes attended)
- Previous Marks (past academic performance)

The processing layer is responsible for:
- Data preprocessing
- Training the machine learning model
- Applying the model to new inputs

Finally, the output layer provides:
- Predicted final marks
- Corresponding grade (A, B, C, etc.)

The system is designed to be user-friendly and efficient, ensuring that even non-technical users can easily interact with it.

4. SYSTEM IMPLEMENTATION

The implementation of the system is carried out using Python, along with powerful libraries such as Pandas and Scikit-learn. Pandas is used for handling and organizing the dataset, while Scikit-learn provides tools for building and training the machine learning model.

The core algorithm used in this project is **Linear Regression**, which establishes a relationship between input variables and the output variable. It predicts the final marks based on how strongly each input factor influences the result.

The implementation process involves:

- Loading the dataset into a DataFrame
- Separating input features and target variable
- Training the model using historical data
- Accepting user input
- Generating predictions

The system processes the input data and applies the trained model to produce accurate predictions. The predicted marks are then converted into grades to make the output more understandable.

5. RESULTS AND DISCUSSION

The system produces reliable predictions based on the given inputs. The results show that factors like study hours, attendance, and previous marks significantly influence student performance.

Some key observations from the results include:

- Students with higher study hours tend to achieve better marks
- Attendance plays a crucial role in performance
- Previous marks strongly affect future predictions

The model provides instant results, making it highly efficient for real-time use. However, the accuracy of the predictions depends on the quality and size of the dataset used for training.

Overall, the system demonstrates that even a simple machine learning model like Linear Regression can provide meaningful insights into student performance.

6. TESTING

Testing is an essential part of the system to ensure its accuracy and reliability. Different types of testing were conducted during the development process.
Unit testing was performed to verify individual components such as data input and prediction functions. Integration testing ensured that all modules work together correctly. System testing was carried out to evaluate the overall performance of the application.

Various test cases were used, including different combinations of input values, to validate the correctness of predictions. The system handled all test cases effectively and produced consistent results.

The testing process confirmed that:

- The system is stable and reliable
- Predictions are consistent
- The model performs well for different inputs

7. CONCLUSION AND FUTURE ENHANCEMENTS

The Student Grade Predictor successfully demonstrates how machine learning can be applied to solve real-world educational problems. The system is simple, efficient, and capable of providing quick predictions based on key academic factors.
In conclusion, the project:

- Predicts student performance effectively
- Provides fast and accurate results
- Helps in early decision-making
- Supports student improvement

Despite its advantages, the system has some limitations. It considers only a few input factors and uses a basic machine learning model. Therefore, there is scope for further improvement.

Future enhancements may include:

- Adding more input parameters such as stress levels, health, and assignments
- Using advanced models like Random Forest or Neural Networks
- Improving prediction accuracy
- Developing a web or mobile-based application
- Integrating data visualization tools

8. REFERENCES & BIBLIOGRAPHY

The development of this project was supported by various standard resources, including:

- Python official documentation
- Pandas library documentation
- Scikit-learn library documentation
- Machine Learning textbooks
- Online tutorials and research papers on student performance prediction